

claude-windows / 985cc286-9e1e-4217-87c3-2d418bc4fa9c

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\985cc286-9e1e-4217-87c3-2d418bc4fa9c\subagents\workflows\wf_cb1d52fa-fed\agent-acc75c95d8ebb31d7.jsonl`  
- SHA-256: `4b7eaadb8834ca26fe1edd2ec763c9430948e885b25a7296455c9ae807110630`  
- Source modified: `2026-06-14T08:26:55+00:00`  
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- project: `wf_cb1d52fa-fed`  
- session_id: `985cc286-9e1e-4217-87c3-2d418bc4fa9c`
```

Transcript

1. user (2026-06-14T08:24:06.513Z)

THE PIVOT under study = "PERSONALIZATION-FIRST": instead of one universal model for all users, ship a BATTERIES-INCLUDED baseline that CONTINUALLY SELF-OPTIMIZES to a SINGLE user's own footage. As the user's corpus grows (10-15 videos), the on-device model + PER-USER CUE SELECTION (driven by the EXISTING LOVO / A-B experiment harness run on the user's OWN videos, with the user's CORRECTION LOOP as the per-user label source) becomes heavily tuned for THAT user ? deep analysis of THEIR rallies + shot types (serves, smashes, drives, drop shots). Per-user decisions like "the motion/person cue gives no F1 win for THIS user -> default it OFF for them." The model is optimized for the user's videos, NOT others (wins may hold anecdotally on other footage but weaker). Periodic "BATTERIES UPGRADES" improve the shipped baseline so it needs merely <5 videos to reach superior quality. Two user archetypes: (a) an economically-generous hobbyist who records sessions some days; (b) a professional with 100+ hours of their own video. CONTRAST = "generalization-first" (build one model that works on everyone's videos). The owner believes the harness + eval + A/B were DESIGNED to enable this roadmap.

DIMENSION: PRODUCT / PMF / market. Analyze: who wants a per-user, on-device "analyze MY own videos deeply" tool vs a generic cloud analyzer; the value prop of "your videos, your model, your insights, on your machine"; the privacy/ownership appeal; pricing (one-time install + a "batteries-upgrade" subscription) for the generous-hobbyist vs the 100+hr professional. Do 2-4 web searches for the competitive landscape of racket-sports / general-sports video analysis tools (e.g. SwingVision, Baseline, Dartfish, Hudl, coach-facing tools) ? are they cloud/generic or local/personal, and what do they charge? Return pros/cons of personalization-first for the PRODUCT, plus how it shapes future product choices.

2. assistant (2026-06-14T08:24:10.096Z)

I'll analyze the "personalization-first" pivot from a product/PMF/market dimension. Let me start by understanding the existing harness/eval/correction-loop infrastructure that this roadmap claims to build on, then research the competitive landscape.

3. user (2026-06-14T08:24:11.024Z)

```
1      # PMF & Market Study ? AI Highlight-Reel Tool for Self-Recorded Racket Sports  
2  
3      > **Research snapshot ? 2026-06-07.** Figures are cited as-of with source + year;  
4      > validate live before relying on them for investment/pricing decisions. This is the  
5      > market-research-of-record behind the PMF section of the living  
6      > [`COMMERCIALIZATION.md`](COMMERCIALIZATION.md) tracker ($6). Product framing:  
7      > a closed-source hosted API/SaaS (plus local mode) that ingests long-form,  
8      > self-recorded, **static-camera, full-court amateur** racket-sports video and  
9      > auto-detects rallies and compiles highlight reels. Primary sport: badminton;  
10     > sport-generic architecture roadmapped to tennis, pickleball, table tennis, cricket.  
11  
12     > **Confidence convention.** Figures from named industry reports, federations, or  
13     > company pricing pages are tagged with source + year. Inferred items are tagged
```

14 > ***(estimate)** / ***(low confidence)****. Emerging-category market sizes vary widely
 15 > between vendors; ranges are shown rather than false precision.
 16
 17 ---
 18
 19 ## 1. Executive summary
 20
 21 The product sits at the intersection of three favorable trends: (a) a fast-growing
 22 **sports analytics market** (~USD 5?6B in 2025, ~18?22% CAGR to USD 24?36B by ~2034
 23 depending on vendor); (b) a structural shift to **amateur self-recording** on action
 24 cameras and phones (action-camera market ~USD 6.5B in 2024, sports the largest
 25 segment); and (c) **strong racket-sport participation** ? 220M+ badminton players,
 26 106M tennis players (2024), and a near-vertical pickleball curve (~19.8M US players in
 27 2024, +45.8% YoY).
 28
 29 The central tension is the **static-camera constraint**: the tool works best on fixed,
 30 full-court amateur footage and ***fails*** on broadcast/panning footage. This is both a
 31 moat (most competitors target team sports / broadcast) and a filter on the addressable
 32 user ? the customer must already set a camera on a tripod behind the court. That
 33 behavior is concentrated in **coaches/academies, clubs, and serious enthusiasts**, not
 34 the median casual player.
 35
 36 **Recommended beachhead (strongest single):** **badminton coaching academies and**
 37 **serious club/competitive players in India** ? a large, rapidly formalizing academy
 38 ecosystem (Khelo India tailwind), genuine fixed-angle recording behavior, existing
 39 willingness to pay for coaching, an underserved tooling gap (incumbents are
 40 team-sports/broadcast), and a direct-but-immature competitor field. The product is
 41 ***native badminton-first*** ? a genuine edge here. Secondary: ***(2) US/global pickleball**
 42 **players & clubs** (capital-rich, fastest-growing, but SwingVision already strong);
 43 ***(3) badminton academies/clubs in East/SE Asia + Denmark** (highest density).
 44
 45 **Pricing** clusters into hardware-bundled auto-filming (Veo, Trace, Pixellot, Spiideo)
 46 and software-only phone AI (SwingVision USD 179.99/yr; BadPro+ pay-per-use). A
 47 software-only BYO-camera model undercuts the hardware players; realistic anchors
 48 ***(estimate): B2C USD 60?180/yr; B2B academy/club USD 300?1,500/yr per location.****
 49
 50 **Top risks:** (1) amateurs won't reliably set up a tripod ? TAM collapses to
 51 coaches/clubs; (2) free/near-free alternatives cap B2C WTP; (3) capture-quality
 52 variance is the core failure mode; (4) badminton's strongest markets (India, Indonesia,
 53 China) are the most price-sensitive and hardest to bill at Western SaaS prices.
 54
 55 ---
 56
 57 ## 2. Market sizing & trends
 58
 59 ### 2.1 Sports analytics / sports-video market
 60 - ***2025 size:** ~USD 5.47B (Precedence Research) to ~USD 5.79B (Fortune Business
 61 Insights); other vendors USD 5.6?6.0B. ***(2025)***
 62 - ***Growth:** CAGR 15.7%?22.5%. Precedence ? ~USD 29.75B by 2034 (20.63%); Fortune BI
 63 ? USD 24.03B by 2032 (22.5%); MarketsandMarkets (narrower) USD 4.75B by 2030 (15.7%);
 64 Future Market Insights ? ~USD 36.2B by 2035 (22.1%). ***(reports 2024?2025)***
 65 - ***Video analytics is the leading segment*** within sports analytics (Grand View /
 66 Fortune BI) ? the directly relevant sub-market.
 67 - ***Interpretation (estimate):** the ***amateur/club auto-highlight*** slice is a small but
 68 fast-growing fraction; most analytics revenue today is pro/broadcast/team-sports. The
 69 product targets the long tail the big numbers under-serve.
 70
 71 ### 2.2 Action-camera / amateur recording adoption
 72 - ***2024 size:** ~USD 6.5B (GMIInsights) ? ~USD 16.9B by 2034 (~10% CAGR). ***(2024?2025)***
 73 - ***Sports is the largest end-use segment*** of action cameras (GMIInsights, 2024).
 74 - ***GoPro ~47% share, DJI ~20%*** (electroIQ/industry, 2024); phones + cheap tripods are
 75 the de-facto amateur rig.
 76 - ***Implication:** the capture-side behavior the product depends on is already
 77 widespread and growing ? ***for the segments that record*** ? validating "BYO-camera,
 78 software-only."**

79

80 ### 2.3 Participation & popularity by sport and geography

81

82 | Sport | Participation signal | Source (year) |

83 |---|---|---|

84 | **Badminton** | 220M+ play regularly; 160+ countries | BWF est. via BadmintonBites (2023) |

85 | Badminton ? China | ~80M registered, up to ~100M (largest base) | industry compilations (2023) |

86 | Badminton ? India | 20M+ recreational by 2023; academy boom + Khelo India | industry compilations (2023) |

87 | Badminton ? Indonesia | National sport; ~1M club players | industry (est.) |

88 | Badminton ? Denmark | 100,000+ registered club members (Europe #1) | industry (2023) |

89 | Badminton ? Japan/Korea | Major school-club + workplace sport | industry (2023) |

90 | **Tennis** | **106M players (2024)**; +25.6% over 5 yrs (87M in 2021) | ITF Global Tennis Report (2021, 2024) |

91 | Tennis ? Asia | ~1/3 of players (33M of 87M) | ITF (2021) |

92 | **Pickleball ? US** | **19.8M players (2024)**; +45.8% YoY, +311% over 3 yrs; ~6.2M core | SFIA / Pickleheads / The Dink (2024) |

93 | Pickleball ? US courts | ~70,000 courts, +55% places to play YoY | SFIA / Pickleheads (2024) |

94 | Pickleball ? global | IFP 78 member countries; UK +73% in 2024; ~282M played monthly per PPA claim **(low confidence)** | various (2024?2025) |

95 | **Table tennis** | 30M+ players; 227 member associations | ITTF (current) |

96 | **Cricket** | 2.5B+ fans; 500M+ in India (mostly *fans*, not self-recorders) | Cricket Rise / Nielsen (2024?2025) |

97

98 - **Pickleball market value:** ~USD 2.2B (2024) ? USD 9.1B by 2034 at 15.3% CAGR

99 (Market.us/Custom Market Insights, 2024?2025).

100 - **Badminton equipment market:** ~USD 7.23B global (2023), Asia-Pacific dominant.

101

102 **Geographic read:** badminton density is overwhelmingly Asian + Denmark; tennis is

103 broad/high-income; pickleball is US-dominant, capital-rich, globalizing fast; cricket

104 is huge as *fandom* but its amateur self-recording + static-full-court fit is weak

105 (large field, not a fixed-frame court). For *this product's static-camera fit*,

106 badminton/pickleball/tennis/table-tennis are the core; cricket is a long-tail roadmap

107 item, not a beachhead.

108

109 ---

110

111 ## 3. Customer segment evaluation

112

113 | Segment | Market-size signal | Willingness to pay | Accessibility | Static-camera fit | B2C/B2B | Verdict |

114 |---|---|---|---|---|---|---|

115 | **Individual amateur/enthusiast** | Largest raw TAM (100s of millions) | **Low** ? competes with free; ~USD 5?15/mo ceiling | High via app stores; high CAC/churn | **Mixed** ? most won't set a tripod | B2C | Volume play, weak economics; use as funnel |

116 | **Coaches & coaching academies** | India academy boom; thousands across Asia | **High** ? already sell coaching; USD 300?1,500/yr plausible | Reachable via federations/distributors/social | **Strong** ? already film fixed-angle | B2B/B2B2C | **Best fit** ? buyer, recorder, payer aligned |

117 | **Clubs & leagues** | Denmark 100k+; thousands of clubs Asia/EU | Medium-high; USD 500?3,000/yr | Medium ? committee sales cycle | **Strong** | B2B/B2B2C | Strong secondary; longer cycle |

118 | **Tournament/event organizers** | Many small/regional events | Medium ? per-event/seasonal | Medium ? episodic | **Strong** | B2B | Good per-event upsell; lumpy |

119 | **Schools & universities** | Large (esp. Asia/Japan clubs) | Low-medium ? budget-constrained | Low ? slow procurement | Strong | B2B | Slow; grant/Khelo-India dependent |

120 | **National/regional federations** | Few, high-value | Medium-high; procurement-heavy/political | Low ? long cycles | Strong | B2B | Lighthouse/credibility, not beachhead |

121 | **Broadcasters / streamers** | High value per deal | High | Low ? incumbents entrenched | **Poor** ? panning = failure mode** | B2B | **Avoid** ? violates core constraint |

122

123 **Key dynamics:** the static-camera constraint is a **B2B filter** ? coaches/academies/

clubs already record from a fixed angle, so the product fits existing behavior (cleanest PMF). B2C is a large low-WTP, high-CAC funnel best used to feed B2B. Broadcasters explicitly violate the constraint ? exclude.

4. Competitive & pricing benchmarks

Product	Category	Model	Price	Notes (year)
Veo (Cam 3)	Team-sports auto-filming	Hardware + sub	Camera ~USD 1,533 (5G ~1,886) + ~£395?895/yr	15,000+ clubs/80 countries; \$115M raised (2025)
Trace	Soccer auto-filming	**Free camera + sub**	USD 180/yr (Basic) ? 300 (Pro)	Camera leased free with sub (2025)
Pixellot (Air)	Auto camera/streaming	Hardware + sub	Camera ~USD 949 + USD 69?167/mo	Institutional/league; 10,000+ clubs (2025)
Spiideo	Auto-filming/analytics	Sub incl. camera	~USD 3,800/yr (1 team) ? 6,100/yr (club)	Institutional budgets (2025)
Hudl	Team video analysis	SaaS tiers	From ~USD 400/yr	Broad team-sports incumbent (2026)
SwingVision	**Tennis/pickleball AI, single iPhone**	Software freemium	**USD 179.99/yr**	free 2 hrs/mo Closest analog; 200k+ monthly users; \$10M+ raised (2025)
BadPro+	**Badminton AI on phone**	**Pay-per-use, no sub**	Per-analysis (? "< one coaching session")	Direct badminton competitor; shot detection, heatmaps, auto-highlights (2025)
	Niche pickleball/badminton apps	Various	Mixed	Varies Fragmented; few true auto-highlight tools (2025)

What buyers actually pay (synthesis): consumer software-only anchors at ~USD 100?200/yr (SwingVision 179.99); hardware-bundled USD 180?900/yr prosumer/club, USD 3,800?6,100/yr institutional; badminton-specific BadPro+ uses *pay-per-use* ? signaling a real badminton AI market but *unproven subscription WTP* (opportunity + warning). **Positioning:** software-only + BYO-camera undercuts hardware players and sells worldwide via SaaS. Realistic anchors *(estimate): B2C USD 60?180/yr; B2B academy/club USD 300?1,500/yr** per location. The differentiators (cheap offline self-improving detection, per-rally stats/tagging, clean licensing) support a **B2B** value story more than a B2C one.

5. Recommended beachhead

Ranked by **WTP × participation density × existing recording behavior × static-camera fit × competitive whitespace.**

#1 (strongest) ? Badminton coaching academies & serious players, **India**
 20M+ recreational players (2023); rapidly **formalizing academy ecosystem** + Khelo India tailwind; academies already record fixed-angle; coaches are buyers who monetize instruction (high relative WTP); incumbents ignore badminton; the only direct badminton AI (BadPro+) is early with unproven pay-per-use. Product is *native badminton-first*. **Watch-outs:** price sensitivity (need INR-tuned tiers); sell to the **academy**, not the individual.

#2 ? Serious pickleball players & clubs, **US (then UK/Australia/Asia)**
 Fastest-growing sport (19.8M US, +45.8% YoY, 2024); capital-rich; ~70,000 courts; strong recording/sharing culture; pickleball is on the roadmap. **Watch-out:** **SwingVision already owns single-camera tennis+pickleball AI** at USD 179.99/yr (200k+ monthly users) ? head-to-head differentiation, not greenfield.

#3 ? Badminton academies/clubs, **East/SE Asia (China, Indonesia, Malaysia, Japan, Korea) + Denmark**
 Highest global density (China 80M+; Denmark 100k+ structured club members; Japan/Korea school-club & workplace systems), mature club infra, fixed-court recording. **Watch-outs:** China needs local hosting/partners + is regulatory-complex; SE Asia price-sensitive; Denmark/EU small but high-WTP and easy to pilot.

```

179
180 **Bottom line:** lead with **India badminton academies** (whitespace + behavior + buyer
181 alignment), use **enthusiast B2C as the funnel**, validate a **club/academy B2B2C
182 motion**, then expand to **pickleball-US** (out-build SwingVision) and
183 **badminton-Asia/Denmark** (density).
184
185 ---
186
187 ## 6. Risks & open questions (what would invalidate the thesis)
188
189 1. **Tripod/static-camera adoption among amateurs.** If serious amateurs won't set up a
190 fixed camera, B2C TAM collapses to coaches/clubs. *Validate:* observe 3?5 academies +
191 50?100 enthusiasts on whether they already record fixed-angle.
192 2. **Free / near-free alternatives** (YouTube, free app tiers, BadPro+ pay-per-use) cap
193 B2C WTP. *Validate:* fake-door pricing at USD 60?180/yr.
194 3. **Capture-quality variance = the failure mode** (gym lighting, off-center angle,
195 occlusion, mixed resolutions). *Validate:* benchmark rally-detection P/R on 100+ real
196 amateur clips; define a "good-capture" spec + auto-reject (ties to
197 `VIDEO_RECORDING_GUIDELINES.md` + the observability reports).
198 4. **Monetization mismatch in best-participation markets** (India/Indonesia/China are
199 price-sensitive, hard to bill). *Validate:* local pricing/payments pilot; per-academy
200 INR licensing.
201 5. **Competitive response** (SwingVision or a team-sports incumbent adds racket sports).
202 *Validate:* ship a badminton-quality advantage incumbents can't trivially match
203 (offline self-improving detection, per-rally tagging).
204 6. **B2B sales-cycle drag** for clubs/federations/schools. *Validate:* single-coach/
205 academy self-serve before institutional deals.
206 7. **Clean-licensing positioning** matters mainly to B2B buyers ? confirm it's a real
207 purchasing criterion via 5?10 buyer interviews.
208
209 **Cheapest next steps:** (a) 10?15 discovery interviews with Indian badminton academy
210 owners/coaches; (b) a capture-quality benchmark on real amateur footage; (c) a fake-door
211 pricing test across B2C (~USD 99/yr) and B2B academy (~USD 600/yr).
212
213 ---
214
215 ## 7. Sources
216
217 - Sports analytics market: Precedence Research; Fortune Business Insights; Grand View
218 Research; MarketsandMarkets; Future Market Insights. *(2024?2025)*
219 - https://www.precedenceresearch.com/sports-analytics-market
220 - https://www.fortunebusinessinsights.com/sports-analytics-market-102217
221 - https://www.grandviewresearch.com/industry-analysis/sports-analytics-market
222 - https://www.marketsandmarkets.com/PressReleases/sports-analytics.asp
223 - https://www.futuremarketinsights.com/reports/sports-analytics-market
224 - Action camera market: https://www.gminsights.com/industry-analysis/action-camera-
225 https://www.precedenceresearch.com/action-camera-market (2024);
226 https://electroi.com/stats/action-camera-statistics/ (2024)
227 - Badminton participation/market: https://badmintonbites.com/how-many-badminton-players-
228 are-there-in-the-world/ ;
229 https://stringsandpaddles.com/where-is-badminton-played-10-countries-where-its-
230 popular/ ;
231 https://en.wikipedia.org/wiki/Badminton ;
232 https://www.marketgrowthreports.com/market-reports/badminton-market-110049 ;
233 https://www.thesportsschool.com/top-10-badminton-academies-in-india/
234 - Tennis: https://www.itftennis.com/en/news-and-media/articles/itf-global-tennis-report-
235 participation-hits-106-million-in-five-years/ ;
236 https://sgbonline.com/itf-global-tennis-report-2021-shows-worldwide-participation-
237 gains/
238 - Pickleball: https://sfia.org/resources/as-pickleball-continues-unprecedented-growth-
239 in-every-age-group-and-region-for-third-straight-year-significant-investments-still-needed-for-
240 court-and-facility-demand/ ;
241 https://www.thedinkpickleball.com/report-pickleball-participation-surges-to-
242 nearly-20-million-in-2024/ ;

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236     https://www.pickleheads.com/guides/pickleball-statistics ;
237     https://www.custommarketinsights.com/report/pickleball-market/ ;
https://market.us/report/pickleball-market/
238     - Table tennis: https://en.wikipedia.org/wiki/International_Table_Tennis_Federation
239     - Cricket: https://cricketrise.com/cricket-fans-in-india/ ;
240     https://www.nielsen.com/insights/2025/global-cricket-connecting-fans-driving-roi/
241     - Competitors & pricing: https://www.véo.com/en-us/pricing ; https://traceup.com/pricing
;
242     https://www.pixellot.tv/ ; https://www.hudl.com/pricing ;
https://swing.vision/subscribe ;
243     https://www.badproplusapp.com/
244

```

4. user (2026-06-14T08:24:11.416Z)

```

1      # Commercialization Tracker
2
3      **Status:** ? LIVING ? track this in parallel with product/quality work.
4      **Owner:** avidullu . **Last refreshed:** 2026-06-07
5
6      **Purpose.** Single source of truth for the **commercial** dimension of the project:
7      licensing/legal posture, codec & output economics, hosting unit-economics, the
8      golden-set vendoring spend, packaging/distribution, and the product-market-fit
9      risks that gate monetization. Updated as decisions land ? see the
10     [Changelog](#changelog) at the bottom.
11
12     **Merged from / supersedes:**
13     - `archives/MONETIZATION_AUDIT.md` (2026-05-30 deep-research licensing & economics
14       audit) ? kept as the **research-of-record** (per-claim verification transcript,
15       primary sources). This doc carries its *current* status forward.
16     - `archives/COMMERCIAL_READINESS_REVIEW.md` (code-health + PMF assessment) ?
17       archived; its commercial findings + the quality-phase priorities live here now.
18
19     **Premise (unchanged, verified):** the product is a **closed-source hosted API** ?
20     users upload videos to a cloud NVIDIA GPU VM and get highlight reels back. We never
21     distribute code/binaries, so GPL/LGPL *distribution* copyleft is not triggered ? but
22     **AGPL §13 (network clause)**, **metered-API terms**, and **video-codec patent
23     royalties** all still apply.
24
25     ---
26
27     ## 1. Commercialization dashboard
28
29     Legend: ? done . ? in progress / watch . ? blocker / needs action . ?? needs legal
30     counsel . ? future
31     Priority: **P0** (do first) . **P1** . **P2** (lowest). Rows are grouped **Pending ? In-
32     flight ? Completed**, each sorted by priority. (C# IDs are stable identifiers ? they don't imply
33     order.)
34
35     | # | Workstream | Pri | Status | Where it stands | Next action / owner decision |
36     |---|---|---|---|---|---|
37     | | **? PENDING (needs action / not started)** | | | |
38     | C5 | **Output codec royalties** | **P0** | ? | Highlights are still encoded **H.264 +
39     AAC** (`libx264` in `backend/pipeline/stitcher/compiler.py`) ? encode-side AVC patent exposure
40     remains. | Switch output to **AV1 + Opus** (royalty-free) on an **Ada-class GPU (L4/L40)** for
41     HW encode. See §4. Decision: provision Ada-class GPU? |
42     | C9 | **Packaging / distribution** | **P0** | ? | No `pyproject.toml`; the diagnostic
43     `setup.py` shadows the build system. | Design in `DEFERRED_HARDENING_PLAN.md` (#7) ? **awaiting
44     your approval** before code. |
45     | C12 | **SaaS foundations (async/auth/tenancy)** | **P1** | ? | Sync `/api/process`
46     blocks on long videos; no auth/tenancy. | Multi-user/platform strategy in
47     `PLATFORM_ARCHITECTURE.md` (pluggable storage/compute, tenancy, orchestration); async slice =
48     `DEFERRED_HARDENING_PLAN.md` (#16). |
49     | | **? IN-FLIGHT (in progress / watch)** | | | |
50     | C4 | **Runtime AI strategy (A/B/C + owned model)** | **P0** | ? | Live = **(C) Gemini

```

2.5 Flash, paid tier**. Long-term = **own a self-hosted generative video-QA model** (fine-tuned open VLM) as primary, **paid Gemini always-ready as fallback** via the provider registry. | Make "provider registry + paid fallback" a core principle; build the owned model **after** the scaffolding/flywheel are solid. See `PLATFORM\_ARCHITECTURE.md` §5A. |

40 | C13 | **Beachhead validation** | **P0** | ? | Market research
 (`PMF\_MARKET\_RESEARCH.md`) recommends **India badminton academies** as the #1 beachhead. Unvalidated. | Cheapest tests: 10?15 academy/coach discovery interviews; rally-detection P/R benchmark on 100+ real amateur clips; fake-door pricing (B2C ~\$99/yr, B2B academy ~\$600/yr). See §6. |

41 | C8 | **Golden-set vendoring spend** | **P1** | ? | ADR-006: **Roora (Bengaluru)** selected for an initial paid pilot; licensed/owned video only (no user uploads to vendors). | Confirm quality-per-dollar from the pilot; then GS-3 `HumanVendorProvider`. See `GOLDEN\_SET\_VENDORING.md`. |

42 | C3 | **Shuttle-tracking weights provenance** | **P1** | ? ?? | WASB-SBDT **code is MIT (clear)**; distributed **checkpoints are academic-data-trained** ? not covered by the MIT code grant. MonoTrack/YOLO-NAS banned (noncommercial). | **Ship-gating** blocker (multiplies per sport). Resolve via an own-trained detector **or** a clean MIT/Apache detector swap ? NOTE: ADR-002 says our temporal labels can't train the detector (needs coordinate labels), so own-weights is a new, un-scoped workstream, NOT delivered by ROADMAP Phase 4. |

43 | C14 | **Owned-model base licensing + training-data policy** | **P1** | ? ?? | Owned video-QA model must build on a **commercial-clean (MIT/Apache-2.0/BSD), video-capable** base ? primary **Qwen3-VL (Apache-2.0)**, fallback **InternVL3 (MIT)**. **Gemma 4 = AMBER / bench-only** (Apache grant likely but Prohibited-Use posture unconfirmed ? counsel needed; not a build target yet). **Reject Gemma 3** (viral derivative clause ? permanent Google PUP), Llama Vision (naming + 700M-MAU), `gpt-oss` (text-only). **Hard rule: never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs** (competing-model clause). The moat is the **consented dataset + eval harness**, not the model. | **MIT/Apache-2.0/BSD** for code+data+weights; **open-model teachers only** ; separate explicit **training opt-in** ; **exclude minors** ; per-clip consent ledger. See `OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` (authoritative). |

44 | C6 | **Inbound HEVC decode** | **P2** | ? ?? | Decoding users' GoPro H.264/**HEVC** uploads is unavoidable; HEVC pools are fragmented + highest-uncertainty. | Low/uncertain residual; confirm decode-side exposure with counsel before launch. |

45 | C7 | **Hosting unit-economics** | **P2** | ? | Per-video cost is *cheap* for Gemini (~\$0.05 input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline costs unmeasured. | **Benchmark \$/video & latency on the target VM** for A/B/C before pricing. See §5. |

46 | C10 | **Value layer (product surface)** | **P2** | ? | Minimal exports shipped (EDL/JSON/CSV + `basic\_stats`, PR #51). | Deepen: per-rally tags/notes that feed golden, per-player/match stats, clip gallery. See §6. |

47 | C11 | **Input-quality enforcement** | **P2** | ? | Per-video observability reports surface a customer verdict + footgun flags incl. audio-present (PR #50). | Turn more `VIDEO\_RECORDING\_GUIDELINES` dimensions into validators/report rules. |

48 | C2 | **supervision` ByteTrack pin\*\* | \*\*P2\*\* | ? | Pinned `<0.30.0` (deprecation ahead). | Migrate before lifting the pin (BACKLOG). Not a legal risk, a maintenance one. |

49 | | **? COMPLETED** | | | |

50 | C1 | **AGPL / copyleft removal** | ? | ? | ADR-005 shipped: AGPL `ultralytics` YOLOv8 replaced with **torchvision Faster R-CNN (BSD) + `supervision` ByteTrack (MIT)**. No AGPL code/weights in the stack. | Done. Keep new deps permissive. |

51

52 ---

53

54 ## 2. Licensing & dependency posture (refreshed risk table)

55

56 Verdicts updated against the current stack (`requirements.txt` + code). Original

57 2026-05-30 verdicts and the 25-claim verification transcript are in

58 `archives/MONETIZATION\_AUDIT.md`.

59

Dependency	License	SaaS obligation?	Current verdict	Notes
--- --- --- --- ---				
~~`ultralytics` + `yolov8n.pt`~~	AGPL-3.0	Yes (§13)	? **REMOVED**	Replaced per ADR-005 ? no longer in the stack.
torchvision detectors (Faster R-CNN/RetinaNet)	BSD-3	No	? LIVE	The chosen replacement; person = COCO class (see `yolo_hybrid.py` `_PERSON_CLASS_ID`).
supervision (ByteTrack)	MIT	No	? LIVE	Pinned `<0.30.0` ? migrate before bump (C2).

```

65 | **WASB-SBDT** (shuttle) | MIT (code) | No | ? code clear | Weights provenance open
(C3). |
66 | **MonoTrack / YOLO-NAS** | noncommercial | ? | ? BANNED | Do not use. |
67 | **Gemini API** (paid tier) | service terms | Paid: no training/review | ? LIVE WATCH |
Must bill the paid tier to keep user video private (C4). |
68 | H.264 / HEVC / AAC codecs | patent pools | royalties on encode/decode | ? ?? |
Separate from ffmpeg's license. Encode-side open (C5); decode-side residual (C6). |
69 | ffmpeg/ffprobe, google-genai, torch, opencv, numpy,
fastapi/uvicorn/pydantic/jinja2/dotenv | LGPL/Apache/BSD/MIT | No | ? CLEAR | Invoked as
subprocess / imported; all permissive. |
70 | **`obsreport`** (reports) | permissive (private) | No | ? soft dep | Pin the
`git+ssh@vX` tag once the repo is published (BACKLOG). |
71
72 **Bottom line:** the two hard blockers from the audit (Ultralytics; banned weights)
73 are resolved/avoided. Remaining commercial-legal items are **codec royalties (C5/C6)**
74 and **weights provenance (C3)**, plus the metered-API privacy posture (C4).
75
76 ---
77
78 ## 3. Runtime AI strategy ? A/B/C
79
80 The semantic rally-boundary step is the one external/metered dependency. Three paths
81 (from the audit), with current standing:
82
83 - ** (A) Eliminate ? offline learned segmenter.** ActionFormer (MIT) / our own learned
84 classifier on `candidate_segments.stats`. **Best steady-state margin** (no per-call
85 cost, no external dep) and matches the Phase-4 roadmap (substrate = WASB
86 shuttle-motion, decided in ADR-004). **Most engineering.** ? *target*
87 - ** (B) Self-host VLM.** Qwen2.5-VL-7B/32B (Apache-2.0) or InternVL2.5-8B (MIT). No
88 per-call fees; pays in GPU VRAM (~16?24 GB for 7B). ?? Qwen 3B is noncommercial.
89 - ** (C) Keep Gemini 2.5 Flash (paid).** Fastest, cheap per video, video-native;
90 ongoing metered cost + external dependency. **? current live path.**
91
92 Hosted alternatives for (C) comparison: Anthropic Claude (no training on
93 inputs/outputs; customer owns outputs; **no native video** ? weaker here);
94 OpenAI (no training by default; ZDR available); Gemini remains strongest video-native.
95
96 ---
97
98 ## 4. Codec / output economics (the live gap ? C5)
99
100 **The trap:** ffmpeg being free (LGPL) does **not** grant the *patent* rights to the
101 codecs it implements. Royalties attach to the act of encode/decode.
102
103 - **Encode side (we control it):** highlights currently emit **H.264/AAC**. AVC (Via LA)
104 is first 100k units/yr royalty-free with a permanent "free to end-users over the
105 internet" provision ? a small SaaS likely falls under thresholds, but ?? confirm.
106 - **Mitigation:** emit **AV1 (SVT-AV1, BSD + AOMedia royalty-free patent grant) + Opus
107 (royalty-free)**. NVENC AV1 HW-encode needs **Ada-class GPUs (L4/L40)**; Ampere/Turing
108 must use **SVT-AV1 (CPU)**.
109 - **Decode side (unavoidable, C6):** inbound GoPro **HEVC** decode ? fragmented pools,
110 highest uncertainty; low residual.
111
112 **Recommendation:** provision an **Ada-class GPU (L4/L40)** and standardize highlight
113 output on **AV1 + Opus** ? removes the encode-side royalty question almost entirely.
114 *Implementation lives in `backend/pipeline/stitcher/compiler.py` (codec args).*
115
116 ---
117
118 ## 5. Hosting unit-economics (C7)
119
120 Reference workload: ~30-min 1080p match on a cloud NVIDIA GPU VM.
121
122 - ** (C) Gemini 2.5 Flash (paid):** input $0.30/1M tokens, output $2.50/1M; video ~263
123 tok/s ? 10-min clip ? 158k input tokens ? **~$0.05 input**. Cheap per unit; the

```


124 economic risk is ****volume × latency****. Flash-Lite is cheaper; Pro ~10×.

125 - ****(B) Self-host Qwen2.5-VL-7B:**** \$0 API; cost = GPU-hours / throughput; needs ?24 GB

126 VRAM (L4/A10). \$/video vs Gemini is ****open ? measure****.

127 - ****(A) Offline ActionFormer/learned:**** cheapest at inference; cost = one-time training

128 + light GPU. ****Best steady-state margin.****

129

130 > The A/B/C \$/video comparison depends on GPU instance + batch throughput ?

131 > ****benchmark on the target VM before pricing the product.****

132

133 ---

134

135 ## 6. Product-market fit & the quality/data phase

136

137 ### Beachhead (from market research, 2026-06-07 ? see

138 [`PMF_MARKET_RESEARCH.md`])(`PMF_MARKET_RESEARCH.md`))

139 The static-camera constraint is a ****B2B filter****: the buyer must already record from a

140 fixed angle, which points away from the median casual player and toward coaches/clubs.

141 Ranked beachheads:

142 1. ****Badminton coaching academies & serious players, India**** ***(strongest)*** ? 20M+

143 players,

144 formalizing academy ecosystem (Khelo India), genuine fixed-angle recording, coaches

145 who already monetize, badminton whitespace vs incumbents. Sell to the ****academy****,

146 not

147 the individual; needs INR-tuned pricing ****and a Kannada/Hindi product**** ? see

148 [`I18N_PLAN.md`])(`I18N_PLAN.md`) (Bangalore-first localization, by design).

149 2. ****US/global pickleball players & clubs**** ? fastest-growing sport (19.8M US, +45.8%

150 YoY)

151 but ****SwingVision already owns single-camera tennis/pickleball AI**** (USD 179.99/yr) ?

152 head-to-head, not greenfield.

153 3. ****Badminton academies/clubs in East/SE Asia + Denmark**** ? highest density; China

154 needs

155 local hosting/partners; Denmark/EU small but high-WTP, easy to pilot.

156

157 Pricing anchors ***(estimate)***: software-only BYO-camera ****B2C USD 60?180/yr****, ****B2B**

158 academy/club USD 300?1,500/yr** per location (undercuts hardware-bundled Veo/Pixellot/

159 Spiideo; comparable to SwingVision software-only).

160

161 ### PMF risks (from the readiness review)

162 - ****Capture quality is the gate.**** Accuracy is capture-conditional; broadcast/handheld

163 footage is a failure mode. The wedge (static, tripod, full-court) is defensible but

164 narrows the addressable market ? enforce/guide it (C11, `VIDEO_RECORDING_GUIDELINES`).

165 - ****Value beyond raw segments is table stakes.**** Time-saved editing alone is weak;

166 per-rally tags, stats, and a correction loop are what make it sticky (C10).

167 - ****Golden labels are the recurring blocker**** across the offline segmenter, audio

168 fusion, and measurable quality ? the highest-leverage investment (C8).

169

170 ### Refined priority order (quality/data phase ? "focus on quality once new videos

171 land")

172 Carried from the readiness review; commercialization items above are tracked in

173 parallel.

174

175 1. ****Golden-set expansion + data-flywheel UX**** (highest with new labeled videos):

176 ingest the new batch, expand calibration/CV/regression baselines, make label

177 contribution + safe retraining easy and non-contaminating (eval/training separation).

178 2. ****Hybrid unification + observability hooks**** (deferred item 7, now higher): extract

179 shared logic from `tracknet_hybrid`/`wasb_hybrid` before heavy AI timing/metrics

land.

3. ****Detector/windowing quality iteration**** with new data: re-tune velocity/merge/

min-duration + `rally_gate`; validate WASB on real video; trial audio fusion

(ADR-007).

4. ****Deeper value layer**** (beyond minimal exports, C10): per-rally tagging/notes ?

`candidate_segments`, per-player/match stats, clip gallery, human-correction loop.

5. ****Input-quality enforcement & guidance polish**** (C11): more guideline dimensions ?

validators/report rules; "your video is good enough" UX.

6. ****Multi-sport maturation + recording best practices.****

```

180     7. **SaaS / multi-user / async foundations** (C12) ? lower until monetization is firm.
181     8. **Polish & distribution** (C9): installer story, first-run path, codec/output,
economics.
182
183     **Recommendation:** with the mid-week labeled-video batch, spend ~80% on #1 + #3
184     (flywheel + quality iteration on real data), then unification (#2) and deeper value
185     (#4).
186     ---
187
188     ## 7. Open questions needing owner / counsel sign-off ??
189
190     1. **Ultralytics Enterprise price** ? only relevant if a permissive detector ever proves
191     insufficient (fallback). Quote-only.
192     2. **Per-checkpoint / dataset licenses** for WASB / TrackNetV3 / any VLM ? train-own or
193     confirm each (C3).
194     3. **Exact Via LA (AVC) / Access Advance (HEVC) rates & free thresholds** ? confirm live
195     figures with counsel before relying on the "small SaaS under threshold" reading
196     (C5/C6).
197     4. **Real per-video GPU cost & latency** of self-hosted VLM vs Gemini paid at expected
198     throughput (C7).
199     5. **Provision Ada-class GPU (L4/L40)** for HW AV1 encode? (C5 ? gates the codec
switch.)
200     ---
201
202     ## Changelog
203
204     - **2026-06-07** ? Added the market-research-of-record (`PMF_MARKET_RESEARCH.md`):
205     ranked beachhead (India badminton academies #1), pricing benchmarks, and a C13
206     "beachhead validation" workstream. Wired the summary into §6.
207     - **2026-06-07** ? Created this living tracker. Merged the 2026-05-30 monetization
208     audit + the commercial-readiness review. **Refreshed since the audit:** C1 AGPL
209     removal now ? shipped (was the #1 blocker); added C5 as a ? (output is still H.264,
210     the audit's AV1 recommendation is not yet implemented); C9 packaging tied to
211     `DEFERRED_HARDENING_PLAN.md`; reflected PR #50/#51 (input-quality reports, value-layer
212     exports). Archived `COMMERCIAL_READINESS_REVIEW.md`.
213
214     ---
215
216     ## See also
217     - `PMF_MARKET_RESEARCH.md` ? market-study-of-record (sizing, segments, competitors,
beachhead; cited).
218     - `archives/MONETIZATION_AUDIT.md` ? licensing/economics research-of-record (sources +
verification transcript).
219     - `archives/COMMERCIAL_READINESS_REVIEW.md` ? superseded assessment (history).
220     - `archives/decisions/DECISIONS.md` ? ADR-005 (permissive stack), ADR-006 (vendor
process).
221     - `ROADMAP.md` ? Phase-4 offline segmenter (the C4 "(A) eliminate" path).
222     - `DEFERRED_HARDENING_PLAN.md` ? packaging (C9), async (C12) designs (pending approval).
223     - `GOLDEN_SET_VENDORING.md` ? vendor process + Roora pilot (C8).
224     - `BACKLOG.md` ? `supervision` migration (C2), `obsreport` pin, codec follow-ups.
225

```

5. assistant (2026-06-14T08:24:14.485Z)

Now let me look at the harness/eval/correction-loop docs that the pivot claims were designed for this, plus the strategy docs, while I kick off web research.

6. user (2026-06-14T08:24:15.451Z)

```

1     # Experiment harness ? A/B + shadow eval for rally-detection quality (no-regression by
design)
2
3     **Status:** **P1 IMPLEMENTED** (`backend/eval/experiment.py` +

```

```

`tests/test_experiment.py`, shipped
4      2026-06-13) ? the `Variant` layer, `compare()` (labeled A/B, LOVO-honest),
`compare_unlabeled()`
5      (SxS on NEW unlabeled footage), and the leaderboard. **P2?P4 remain owner-gated.**
Phases below are
6      sequenced but each is a separate PR.
7
8      > **Companion docs:** the cues this experiments *on* are in
9      > [MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md); the scoring mechanics are
10     > [EVALUATION.md](EVALUATION.md); the no-regression gate + golden labels +
`baseline.json` come from
11     > [GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md); the modular-cue architecture is
**ADR-007**
12     > ([archives/decisions/DECISIONS.md](archives/decisions/DECISIONS.md)); the distilled
classifier is
13     > **ADR-004**. The 2-layer mental model is
[HOW_RALLY_DETECTION_WORKS.md](HOW_RALLY_DETECTION_WORKS.md).
14
15     ---
16
17     ## 1. Context ? why this, why now
18
19     PR #112 (merged) added the multi-signal fusion design ? shuttle + person + audio cues,
all **default
20     OFF**. The owner's directive for *how* we evolve quality from here:
21
22     > "Evolve the codebase so we can keep finding new ways to improve quality and
experiment with them ?
23     > or add them as a signal ? without suffering any regressions if the idea doesn't work
out. Rolling
24     > back the code would be too dangerous."
25
26     So **experimentation and deployment must be decoupled.** The contract:
27
28     1. Merge experimental code freely ? it's **gated OFF**, so it cannot change production
behavior.
29     2. Test it **offline against golden labels** (most experiments never touch the served
path at all).
30     3. Promote a variant to default **only on measured lift**, through a CI eval gate.
31     4. **"Rollback" = flip a config flag, never `git revert`.** The proven path is a pinned
variant that is
32         always present.
33
34     **The pleasant surprise (Explore audit, 2026-06-13):** ~80% of this harness already
exists and is
35     simply not composed *as a system*. The missing piece is a thin **variant/experiment
layer + the
36     discipline**, not a platform. §4 is the exact code surface so the next session does not
re-discover it.
37
38     ---
39
40     ## 2. The thesis ? separate expensive DETECTION from cheap DECISION
41
42     The expensive, GPU/WSL-bound, risky part is **detection**: "what signals exist per
frame" (shuttle
43     WASB trajectory, person tracks/boxes, audio hits). The cheap, swappable, pure-Python
part is the
44     **decision**: "given those signals, where are the rallies" (windowing + gates + the
classifier).
45
46     > **Run detection once per video, persist all per-cue signals + per-candidate features,
then re-score
47     > any number of variants OFFLINE ? deterministically, with no GPU and zero production
risk.**

```

```

48
49     This is not aspirational ? `distill_local.py` and `calibrate_local.py` already do
exactly this on
50     stored WASB trajectories. The candidate-segments table (`stats` JSON) is the persistence
substrate.
51     ? Most experiments are pure re-scoring of stored data; they never enter the served
pipeline.
52
53     ---
54
55     ## 3. What already exists (the audit ? do NOT rebuild)
56
57     Exact public surface, with `file:line`. **This table is the anti-duplicate-work
artifact** ? start here.
58
59     ### 3.1 Scoring (variant-agnostic ? grades any `List[Interval]`)
60     `backend/eval/metrics.py`
61     - `interval_iou(a, b) -> float` (:15) ? temporal IoU of two `(start,end)` intervals.
62     - `match_intervals(preds, gts, iou_threshold=0.5) -> MatchResult` (:48) ? greedy 1-to-1
IoU matching.
63     - `score(preds, gts, iou_threshold=0.5, match_result=None) -> Score` (:104) ? P/R/F1,
mean IoU, boundary errors; `Score.as_dict()`.
64     - `pr_curve(preds, gts, thresholds=None) -> List[Score]` (:138).
65
66     ### 3.2 A/B + cross-validation primitives (the comparison engine already exists)
67     `backend/eval/calibration.py`
68     - `improvement_report(predict_fn, baseline_config, tuned_config, gts, iou_threshold=0.5)
-> dict` (:148) ? **before/after metrics + %?. This IS the A/B delta.**
69     - `leave_one_video_out(videos, configs, metric="f1", iou_threshold=0.5) -> dict` (:113)
? **LOVO: pick on other videos, score on held-out video (honest cross-video).**
70     - `cross_validate(predict_fn, configs, gts, k=5, metric="recall", iou_threshold=0.5) ->
dict` (:72) ? within-video k-fold overfit guard.
71     - `select_best(predict_fn, configs, gts, metric="f1", iou_threshold=0.5) -> (Config,
float)` (:61).
72     - `evaluate(preds, gts, iou_threshold=0.5) -> Score` (:41); `kfold_indices(n, k)` (:46).
73     - LOVO video dict shape: `{"name": str, "predict_fn": PredictFn, "gts": [Interval]}`.
74
75     ### 3.3 No-regression gate + baselines (the promotion gate already exists)
76     `backend/eval/regression.py`
77     - `regress(db, video_id, ground_truth, stage="final", iou_threshold=0.5, detector=None,
tolerance=DEFAULT_TOLERANCE, baseline_path=DEFAULT_BASELINE_PATH) -> RegressionResult` (:104) ?
score stored preds vs GT, flag if P **or** R drops > tolerance.
78     - `regress_with_provider(...)` (:160) ? same, fetching GT via the annotation provider.
79     - `save_baseline(video_id, stage, precision, recall, f1, ..., gt_fingerprint=None)`
(:80). `load_baselines(path) -> dict` (:68).
80     - `RegressionResult` (:43): `regressed`, `reasons`, `gt_hash`, baseline P/R,
`tolerance`, `has_baseline`; `as_dict()`.
81     - Default baseline file: `eval_baselines/regression_baseline.json` (:33).
82
83     `run_regression.py` ? CI-gatable CLI, `main(argv)` (:52). **Exit codes: 0 ok / 1
regression / 2 no-DB / 3 no-baseline.** Flags: `--video-id --tier --annotations
--stage{candidates,final} --detector --iou --tolerance --baseline --update-baseline --allow-
missing-baseline --json`.
84
85     ### 3.4 Pinnable model artifact
86     `backend/eval/classifier.py` ? `LogisticRegression` (pure numpy, L2):
`fit(X,y,feature_names=,class_weight="balanced")` (:40), `predict_proba` (:75),
`predict(threshold=0.5)` (:81), `to_dict`/`from_dict` (:86/:97), `save`/`load` (JSON)
(:107/:111). **A trained model = one JSON file ? a pinnable, hashable variant artifact.**
87
88     ### 3.5 Offline re-scoring substrate (persist once, re-score forever)
89     `backend/infrastructure/database.py`
90     - `add_candidate_segment(video_id, detector, start_time, end_time, chunk_index=None,
confidence=None, stats=None, detection_params=None) -> Optional[int]` (:240) ?
`stats`/`detection_params` stored as JSON; `candidate_segments` table cols incl. `stats`,

```

```

`detection_params`, `human_label`, `confirmed_segment_id` (schema :97).
91     - `query_candidate_segments(video_id, detector=None, only_unlabeled=False) ->
List[dict]` (:277) ? returns rows with deserialized `stats`.
92
93     ### 3.6 Label assignment + feature glue
94     `backend/eval/training.py`
95     - `label_candidates(candidate_intervals, gt_intervals, iou_threshold=0.5) -> List[int]`
(:50) ? y-labels by IoU match to golden (same matcher as the evaluator).
96     - `build_training_set(candidate_rows, gt_intervals, iou_threshold=0.5,
feature_fn=extract_yolo_features) -> (X, y, intervals)` (:64) ? pluggable `feature_fn`.
97
98     ### 3.7 Existing LOVO drivers to mirror (not duplicate)
99     - `backend/eval/distill_local.py` ? `run(specs, iou=0.5, threshold=0.5, model_out=None)`
(:98), CLI `python -m backend.eval.distill_local --manifest videos.json --model-out ...`;
`FEATURE_NAMES = [duration, peak_velocity, mean_velocity, active_ratio, net_crossings]` (:40);
already does LOVO over Gemini silver labels.
100    - `backend/eval/calibrate_local.py` ? `run(specs, iou=0.5, metric="f1")` (:77);
`LocalConfig = (velocity_thresh, merge_gap, min_window_duration, min_crossings)`; grid + LOVO.
101    - `backend/eval/calibrate_wasb.py` ? `run(...)` , `load_golden_gts()`;
`backend/eval/gt_loader.py::load_gt_intervals(csv_path, *, strict=True)` is THE canonical golden
reader.
102
103    ### 3.8 Registries + config knobs (variant selection)
104    - `backend/pipeline/segmenters/base.py` ? `SegmenterRegistry` (:35), singleton
`segmenter_registry` (:63); registered: `motion`, `gemini`, `yolo_hybrid`, `tracknet_hybrid`,
`wasb_hybrid`.
105    - `backend/annotations/base.py` ? `AnnotationProviderRegistry`, singleton
`annotation_registry`; providers `file`, `auto`.
106    - `backend/config/models.py::IndexingConfig` ? `default_segmenter` (:65, default
`"yolo_hybrid"`), `detector_model` (:72), `detector_score_threshold` (:73),
`yolo_velocity_threshold` (:74), `yolo_frame_skip` (:75), `min_segment_duration` (:68),
`dead_time_merge_gap` (:69), `motion_threshold` (:67). **No `min_crossings` knob yet** (see
fusion P2).
107
108    ### 3.9 Confirmed ABSENT (no duplication risk)
109    Explore found no existing experiment / variant / shadow / A-B machinery ? only a
stray "A/B test"
110    aspiration comment in `backend/pipeline/detectors/base.py` and an "experiment E1" note
in
111    `AUDIO_RALLY_DETECTION_PLAN.md`. The `regress()` + baseline path is the canonical
comparison mechanism.
112
113    ---
114
115    ## 4. The variant abstraction (the one thin thing to add)
116
117    A Variant = a declarative recipe, not a code branch:
118
119    ```
120    Variant = (segmenter_name, IndexingConfig overrides, enabled cues + per-cue params,
classifier model path/hash)
121    ```
122
123    JSON-serializable; selected via the existing `segmenter_registry` +
`IndexingConfig.default_segmenter`.
124    Control = a pinned, frozen variant (recipe + model hash) that is always registered
and is the
125    default. Adding a new cue = a new Variant differing by one flag ? the control recipe is
untouched.
126
127    ---
128
129    ## 5. Three experiment modes
130
131    ### 5.1 Offline A/B (primary, label-driven, zero production risk)

```

```

132     `compare(videos, variant_A, variant_B)` ? a thin driver composing existing
functions:
133     `query_candidate_segments()` (load persisted features) ? re-score each variant ?
`calibration.improvement_report()`
134     + `calibration.leave_one_video_out()` + `metrics.score()`, graded vs golden
(`gt_loader.load_gt_intervals`).
135     Reports per-video and LOVO-honest aggregate IoU/P/R/F1 deltas. This is the A/B
test, and it needs
136     almost no new scoring code ? only the glue.
137
138     Honest reporting (default ON, additive): `compare(..., honest_stats=True)` also
attaches a "honest"
139     block ? per-metric bootstrap CIs, a paired F1 CI + exact sign test (C1), and
F1@k +
140     segment-count ratio (C4) ? alongside the bare-mean fields (existing output
unchanged; honest_stats=False
141     restores the legacy shape). A bare LOVO mean at n=6 is not promotable on its own; the
lift is real when the
142     F1 CI clears 0 and the sign test agrees. Wiring of the course corrections in
143     [ANALYZERS_AND_RECONCILERS.md](ANALYZERS_AND_RECONCILERS.md) §13/C1+C4 (helpers:
`backend/eval/eval_stats.py`,
144     `segmentation_metrics.py`); record_experiment carries the honest F1 into the
leaderboard.
145
146     5.2 Shadow (for cues that touch detection, e.g. the person detector)
147     The new cue runs and persists its signal into the candidate
`stats`/`detection_params`, but the
148     served decision stays control. Offline you then ask "what would variant B have
scored?" via §5.1 ?
149     B never affects output until it wins. This is how a detection-touching change (not just
a re-scoring
150     change) earns its way in without risk.
151
152     5.3 Promotion gate (no-regression, CI-enforced)
153     regression.regress() + baseline.json + run_regression.py exit codes. A variant
becomes the new
154     default only if it doesn't regress the golden set beyond tolerance (exit 1
blocks). Control is
155     pinned ? rollback = flip default_segementer/variant config, never a code revert.
156
157     ---
158
159     6. No-regression guarantees (the heart of the owner's ask)
160
161     - New cues/signals default OFF ? per-cue enable flag in IndexingConfig; merged
code can't change
162     behavior until explicitly enabled.
163     - Control variant pinned ? frozen recipe + classifier model hash, always registered
& default.
164     - Promotion only via the eval gate ? run_regression.py in CI (exit 1 blocks); no
silent default change.
165     - Experiments are records ? variant-spec hash ? per-video + LOVO scores + label-set
hash + timestamp,
166     persisted to a small JSON experiment leaderboard (generalizes baseline.json).
Answers "which
167     signals ever helped, on which video slices," and makes every result reproducible.
168     - Decoupled events ? "merge an experiment" and "change the default" are separate,
independently
169     revertible actions.
170
171     ---
172
173     7. What to build later (gap list ? thin, additive, owner-gated)
174
175     1. backend/eval/experiment.py ? a Variant dataclass + compare(videos, a, b)

```

```

composing the $3
176     functions (no new scoring math).~~ **DONE (2026-06-13).** Shipped `Variant` (+ pinned
`CONTROL`),
177     `compare` (labeled A/B with LOVO-honest learned-variant training, mirroring
`distill_local`),
178     `compare_unlabeled` (the **SxS-on-new-videos** mode ? agreed / A-only / B-only via
`match_intervals`,
179     no GT), `record_experiment` (? `eval_baselines/experiment_leaderboard.json`), and
`load_video_candidates`
180     (the `query_candidate_segments` bridge). JSON variant recipes first as recommended; a
`VariantRegistry`
181     only if the count grows. 19 tests incl. the no-regression invariant.
182     2. **Persist per-cue features** into candidate `stats` (person: `players_in_court`,
`two_sided_motion`,
183     `in_court_ratio`, `shuttle_player_colocation`; shuttle-state: `net_crossings`,
`direction_reversals`;
184     audio: hit times) so any future variant re-scores offline ? ties to
`MULTI_SIGNAL_FUSION_PLAN.md` $3.2/$4.
185     3. **Per-cue enable flags** in `IndexingConfig` (default `False`); `default_segmenter`
already exists.
186     Add the `min_crossings` knob (fusion P2) here too.
187     4. **CI promotion lane** using `run_regression.py` against the golden `baseline.json`;
write the
188     experiment-leaderboard JSON.
189
190     ---
191
192     ## 8. Phasing (owner-gated; ONE PR per slice, off `master`)
193
194     - **P0 ? this doc**, indexed in `docs/README.md`, cross-linked from
`MULTI_SIGNAL_FUSION_PLAN.md` and
195     `GOLDEN_SET_VENDORING.md`. *(this deliverable; docs-only, no code.)*
196     - **P1 ? `experiment.py` `compare()`** ? **DONE (2026-06-13):** thin wrapper over
`calibration` +
197     `metrics` + `query_candidate_segments`; pure-logic, fully tested in this dev container
(`compare`,
198     `compare_unlabeled`, `record_experiment`, the no-regression invariant).
`compare_unlabeled` adds the
199     SxS read on **new unlabeled footage** the owner asked for (run control + candidate on
a fresh video,
200     diff where they agree/diverge ? two reels via the existing `VideoCompiler` is the GPU-
side follow-on).
201     - **P2 ? per-cue enable flags + feature persistence** into candidate `stats` (default
OFF) ? couples
202     with fusion P1/P2.
203     - **P3 ? CI promotion gate**: `run_regression.py` lane on the golden `baseline.json`;
experiment-leaderboard JSON.
204
205     ---
206
207     ## 9. Scope guardrail
208
209     **Single-tenant / local-first** (per `PLATFORM_ARCHITECTURE.md`): a JSON experiment
registry + the
210     existing harness ? explicitly **NOT** MLflow / W&B / a service. Keep it the size of
`baseline.json`.
211
212     ---
213
214     ## 10. Verification
215
216     - **Offline, runnable in this dev container (no GPU/torch needed for the glue):** unit
test that
217     `compare()` is deterministic on synthetic candidate rows; a **no-regression invariant
test** asserting

```

```
218         "all new-cue flags OFF ? predictions byte-identical to control"; a CI-gate test
asserting
219         `run_regression.py` exit codes (0 ok / 1 regress / 3 no-baseline).
220     - **Machine-side (owner, GPU/WSL):** persist candidates once on a real clip, run
`compare(control,
221         fusion)` LOVO, confirm the reported lift matches a manual `metrics.score()` spot-
check.
222     - **Docs (P0):** links resolve; indexed in `docs/README.md`; cross-linked from
223         `MULTI_SIGNAL_FUSION_PLAN.md` and `GOLDEN_SET_VENDORING.md`.
224     </content>
225
```

7. assistant (2026-06-14T08:24:19.281Z)

I have a strong grasp of the existing PMF research and harness. The existing research is explicitly framed around a ****generalization-first, cloud-hosted API**** model with a B2B (academies) beachhead. The pivot inverts this. Let me run web searches on the competitive landscape to ground the personalization-first analysis.

8. user (2026-06-14T08:24:29.260Z)

Web search results for query: "SwingVision tennis pickleball AI app pricing 2025 cloud or on-device analysis"

Links: [{"title":"SwingVision: Tennis Pick... ? Review 2026: Sentiment & Intel","url":"https://marlvel.ai/intel-report/sports/com-mangolytics-swing"}, {"title":"SwingVision: Tennis Pickleball - App Store - Apple","url":"https://apps.apple.com/us/app/swingvision-tennis-pickleball/id989461317"}, {"title":"SwingVision: Tennis Pickleball App - App Store","url":"https://apps.apple.com/es/app/swingvision-tennis-pickleball/id989461317?l=en-GB&platform=vision"}, {"title":"SwingVision: Tennis Pickleball on the App Store","url":"https://apps.apple.com/by/app/swingvision-tennis-pickleball/id989461317"}, {"title":"SwingVision: AI-powered app for automated tennis & pickleball scoring, stats, highlights, and line calling","url":"https://www.toolify.ai/tool/swingvision"}, {"title":"SwingVision: Tennis & Pickleball AI Stats","url":"https://appshunter.io/ios/app/989461317"}, {"title":"SwingVision Pro Plans & Pricing | Tennis & Pickleball App","url":"https://swing.vision/subscribe"}, {"title":"SwingVision: AI Stats for Tennis & Pickleball ? Liza West","url":"https://www.liza-west.com/deals-discounts/swingvision"}, {"title":"SwingVision: Tennis Pickleball on the App Store - Apple","url":"https://apps.apple.com/eg/app/swingvision-tennis-pickleball/id989461317"}]

Based on the search results, here's what I found about SwingVision:

Pricing

SwingVision Pro costs \$14.99 USD per month or \$179.99 USD per year, with prices varying by country.

Features & Technology

SwingVision is a sports analytics app for tennis and pickleball players that uses AI to provide automated line calling, match highlights, and performance statistics. The app offers features like AI Shot Stats, AI Rally Highlights, Cloud Storage, HD Video, AI Shot Highlights, Line Challenges, AI Scoring, AI Coaching, Family Sharing, 4K Video, and Audio Line Calls.

Analysis Type

The search results mention that the app offers Cloud Storage, suggesting it uses cloud-based processing. However, the search results don't provide explicit details about whether the analysis is performed on-device or in the cloud, or a combination of both. The app does process video data using its AI to automatically generate stats, highlights, and line calls based on your recordings.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. user (2026-06-14T08:24:43.138Z)

Web search results for query: "racket sports video analysis tools comparison Baseline Dartfish Hudl personal player vs coach cloud"

Links: [{"title": "Best Sports Video Analysis Software | 2026 Rankings", "url": "https://worldmetrics.org/best/sports-video-analysis-software/"}, {"title": "Ice Hockey Video Analysis Software: Best Picks (2026)", "url": "https://wifitalents.com/best/ice-hockey-video-analysis-software/"}, {"title": "Top 10 Best Cricket Video Analysis Software | Top Picks 2026", "url": "https://zipdo.co/best/cricket-video-analysis-software/"}, {"title": "A Buyer's Guide to Sport Video Analysis Apps and Software Programs - SimpliFaster", "url": "https://simplifaster.com/articles/buyers-guide-sport-video-analysis/"}, {"title": "Dartfish vs Hudl Sportscode Comparison (2026) | ToolsInfo.com", "url": "https://toolsinfo.com/compare/dartfish-vs-hudl-sportscode/"}, {"title": "myDartfish Pro: the Best Video Analysis Solution in the Market", "url": "https://www.dartfish.com/pro/"}, {"title": "Dartfish vs Hudl Comparison (2026) | ToolsInfo.com", "url": "https://toolsinfo.com/compare/dartfish-vs-hudl/"}, {"title": "A Coach's Guide to Video Analysis Software: Hudl, Dartfish & Top Alternatives", "url": "https://blog.callplaybook.com/blog/coach-video-review-software-hudl-dartfish-alternatives/"}, {"title": "Compare Video Analysis Software | Compare Sports Tech", "url": "https://www.comparesportstech.com/compare-video-analysis-software"}]

Based on the search results, here's a comprehensive overview of racket sports video analysis tools:

Major Platforms Compared

Dartfish and **Hudl** are the two industry benchmarks for sports video analysis. When coaches evaluate video software, these two names stand out as the industry benchmarks, with Hudl being the market leader especially in American team sports.

Dartfish

Dartfish is a pioneer in motion analysis and the preferred tool for individual sports, particularly suited for in-depth biomechanical breakdown in sports like swimming, golf, track and field, and gymnastics where precise form is critical. It supports multi-camera review, time-stamped event tagging, and side-by-side comparisons to make technique differences visible.

Hudl

Hudl ranks first because it combines structured clip workflows with team-wide collaboration, including shared video tagging and coach notes inside review sessions. It's best for game film exchange between teams, creating highlight reels for recruiting, and robust team-wide analysis, and is the industry standard for many high school and college leagues with features well-suited for complex team strategy.

General Comparison Features

Video analysis software includes Hudl, Dartfish, Coach Paint, Ubersense, and Sportscode, with each tool handling tagging and annotation, replay and playback controls, coding and export options, and team or coach collaboration features differently.

The search results focus primarily on these established platforms rather than Baseline specifically or detailed comparisons between personal player vs. coach vs. cloud deployment models for racket sports.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

10. assistant (2026-06-14T08:24:48.570Z)

Let me dig into two more specific angles: on-device/edge AI sports trends and the privacy/ownership positioning, plus the professional-tier video analysis offerings.

11. user (2026-06-14T08:25:00.402Z)

Web search results for query: "badminton AI app personal player analysis your own footage shot detection 2025 2026"

Links: [{"title": "AI-powered Badminton Video Detection: Enhancing Gameplay Analysis and Training", "url": "https://scite.ai/reports/ai-powered-badminton-video-detection-enhancing-A3wQd0Ea"}, {"title": "AI-powered Badminton Video Detection: Enhancing Gameplay Analysis and Training | Opast Publishing Group", "url": "https://www.opastpublishers.com/peer-review/aipowered-badminton-video-detection-enhancing-gameplay-analysis-and-training-5961.html"}, {"title": "AI-powered Badminton Video Detection: Enhancing Gameplay Analysis and Training | TechRxiv", "url": "https://www.authorea.com/doi/full/10.36227/techrxiv.23708325"}, {"title": "AI-Driven Badminton Analysis: Enhancing Shot Detection and Player Performance Strategies | Technology", "url": "https://www.devdiscourse.com/article/technology/3009448-ai-driven-badminton-analysis-enhancing-shot-detection-and-player-performance-strategies"}, {"title": "AI-Powered Badminton Coach: Automated Match Analysis ...", "url": "https://ieeexplore.ieee.org/abstract/document/11144996/"}, {"title": "Journal of Robotics and Automation Research", "url": "https://www.opastpublishers.com/open-access-articles/aipowered-badminton-video-detection-enhancing-gameplay-analysis-and-training.pdf"}, {"title": "MATCHMOTION: AN AI-POWERED MOBILE APPLICATION FOR AUTOMATED BADMINTON VIDEO ANALYSIS USING YOLOV8 OBJECT DETECTION", "url": "https://www.researchgate.net/publication/404597971_MATCHMOTION_AN_AI-POWERED_MOBILE_APPLICATION_FOR_AUTOMATED_BADMINTON_VIDEO_ANALYSIS_USING_YOLOV8_OBJECT_DETECTION"}, {"title": "Custom AI Badminton Video Analysis Software Development", "url": "https://tezeract.ai/industries/badminton-video-analysis-software/"}, {"title": "Badminton: AI-Powered Analysis of Player Patterns", "url": "https://scienceacademy.sph.ac.id/project/badminton:-ai-powered-analysis-of-player-patterns"}]

Based on the search results, here's what I found about badminton AI apps for personal player analysis:

AI-Powered Badminton Analysis Technology

AI-powered badminton video analysis enhances gameplay analysis and training by utilizing artificial intelligence and machine learning algorithms to analyze aspects of badminton game footage, including player movements, shot selection, and game strategy. It provides personalized feedback and recommendations for improvement to players and helps coaches identify patterns and trends in their players' performance.

Practical Applications

MatchMotion is a cross-platform mobile application that enables users to upload badminton match videos for automated AI-powered analysis, employing three specialized YOLOv8 object detection models to identify players, shuttlecocks, and court boundaries within video frames. Custom algorithms analyze detection results to extract game statistics, including rally length, hit counts, shuttle landing positions, and rally winners.

The system can be implemented with existing badminton footage without requiring additional hardware, making it a cost-effective solution for teams and organizations looking to enhance their game analysis capabilities.

Technical Approach

Lightweight models, such as YoloV5, are essential for real-time video analysis, including MobileNets, EfficientNets, SqueezeNet, ShuffleNet, and Tiny-YOLO, designed to be lightweight and optimized for speed while maintaining high accuracy.

AI models leverage advanced pattern recognition and deep learning to classify each shot type

accurately, enabling coaches to analyze technical strengths and weaknesses precisely.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

12. user (2026-06-14T08:25:00.762Z)

Web search results for query: "on-device edge AI sports video analysis privacy local processing no cloud upload 2025 personal training"

Links: [{"title": "How Edge AI Enables Real-Time Video Processing in Smart Cameras - TechNexion", "url": "https://www.technexion.com/resources/how-edge-ai-enables-real-time-video-processing-in-smart-cameras/"}, {"title": "What Is Edge AI? | IBM", "url": "https://www.ibm.com/think/topics/edge-ai"}, {"title": "Edge AI for Internet of Energy: Challenges and Perspectives", "url": "https://arxiv.org/pdf/2311.16851"}, {"title": "Best Mobile Object Detection Models for Edge AI", "url": "https://blog.roboflow.com/mobile-object-detection-models/"}, {"title": "On-Device AI and Edge 2025: The Revolution of Cloud-Free Local Processing", "url": "https://blog.madrigan.com/en/blog/202605131630/"}, {"title": "Privacy-preserving AI video surveillance with edge intelligence - Fluendo", "url": "https://fluendo.com/blog/privacy-preserving-ai-video-surveillance-with-edge-intelligence/"}, {"title": "Benefits of Edge AI for Real Time Surveillance Analytics with Faster Smarter Video Intelligence | Sighthound", "url": "https://www.sighthound.com/blog/benefits-of-edge-ai-for-real-time-surveillance-analytics"}, {"title": "From Factories to Farms, Seven Edge AI Use Cases Powering Real Life - Arm Newsroom", "url": "https://newsroom.arm.com/blog/seven-edge-ai-use-cases-powering-real-life"}, {"title": "Low-power, local AI inference on the edge for a wide range of devices? Here's how it's possible - Edge Up", "url": "https://edgeup.asus.com/2026/low-power-local-ai-inference-on-the-edge-for-a-wide-range-of-devices-heres-how-its-possible/"}]

Based on the web search results, here's what I found about on-device edge AI for sports video analysis with privacy and local processing:

Key Benefits of On-Device Sports Video Analysis

Sports broadcasting is leveraging real-time video analytics to track player movements and provide instant performance insights, and edge AI enables smart cameras to process video locally by minimizing latency and keeping data secure on-site, delivering faster insights and ensuring greater privacy.

Privacy and Security Advantages

With on-device processing, images never leave the device, protecting sensitive environments. This is particularly relevant for personal training applications where athletes may not want video data uploaded to cloud servers. Edge AI helps deliver low-latency, heightened security and cost reduction.

Real-Time Performance for Training Analysis

On-device inference happens in milliseconds, enabling continuous video analysis or frame-by-frame detection, making it ideal for analyzing athletic movements and form in real-time during personal training sessions.

Market Growth

In 2025, the global edge AI market was valued at USD 24.91 billion and is expected to reach USD 118.69 billion by 2033, growing at a CAGR of 21.7% from 2026-2033, indicating significant investment and development in this technology space.

The results show that edge AI technology is well-suited for privacy-conscious sports video analysis applications where data never needs to leave the local device.

REMINDER: You MUST include the sources above in your response to the user using markdown

hyperlinks.

13. assistant (2026-06-14T08:25:04.016Z)

I have enough competitive grounding. Let me do one final search on the personalization/per-user fine-tuning value prop and pricing models (one-time + subscription) to round out the pricing analysis.

14. user (2026-06-14T08:25:12.753Z)

Web search results for query: "software pricing one-time purchase plus subscription updates
"lifetime license" prosumer creative tools 2025 hobbyist vs professional"

Links: [{"title": "Break Free From Adobe: Best Subscription-Free Photo Tools | Fstoppers", "url": "https://fstoppers.com/software/how-break-adobe-2026-subscription-free-creative-suite-719259"}, {"title": "The Future of Software Licensing: Subscription vs. One-Time Purchase in 2025 ? Shop de Arturo", "url": "https://arturo.winandoffice.com/the-future-of-software-licensing-subscription-vs-one-time-purchase-in-2025/"}, {"title": "Software Subscriptions vs Licenses: Which Model Fits You Best?", "url": "https://licensemanager.at/software-subscriptions-vs-licenses/"}, {"title": "Subscription Fatigue in 2025: New Pricing Models & Solutions", "url": "https://www.influencers-time.com/tackling-subscription-fatigue-in-2025-new-pricing-models/"}, {"title": "?2026?Adobe Premiere Pro One-Time Purchase, Lifetime License - Design Offset", "url": "https://design-offset.com/en/adobe-premiere-pro-lifetime-plan/"}, {"title": "?2026?Adobe Photoshop One-Time Purchase, Lifetime License - Design Offset", "url": "https://design-offset.com/en/adobe-photoshop-lifetime-plan/"}, {"title": "?2026?Adobe Lightroom One-Time Purchase, Lifetime License - Design Offset", "url": "https://design-offset.com/en/adobe-lightroom-lifetime-plan/"}, {"title": "Adobe's Creative Software Subscription Model Explained", "url": "https://sparkco.ai/blog/adobes-creative-software-subscription-model-explained"}]

Here are the search results for your query about software pricing models for creative tools in 2025:

Key Findings on Software Pricing Models

****Lifetime/Perpetual Licenses:****

Capture One's perpetual license is listed at \$329.00 as a one-time purchase, and perpetual licenses grant lifetime use of the purchased version, though feature upgrades are tied to new versions and require paid upgrades. Pixelmator Pro is available for approximately \$49.99 on the App Store as a standalone purchase.

****Adobe's Subscription-Only Model:****

A lifetime license for Adobe Premiere Pro is not available, with all current plans exclusively subscription-based in Creative Cloud. The same applies to Adobe Photoshop and Lightroom, which are exclusively subscription-based. Adobe's transition to subscription model in 2013 marked a shift from perpetual licensing to recurring revenue.

****Cost Comparison:****

While subscriptions seem cheaper monthly, over several years they can surpass the cost of a perpetual license. Licenses are more expensive upfront but often cheaper in the long run, though updates usually require additional purchases or fees.

****2025 Pricing Trends:****

Subscription fatigue is reshaping purchasing decisions in 2025 because customers want fewer recurring obligations and clearer value, with the one-time buy returning as a practical alternative. A one-time price is easier to budget than an indefinite commitment, especially for freelancers, small teams, and households.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

15. assistant (2026-06-14T08:25:21.191Z)

I now have enough to synthesize the product/PMF analysis. Let me compose the structured findings on the personalization-first pivot.

Key synthesis points:

- The existing PMF research is built on a **cloud-hosted, generalization-first, B2B-academy** thesis. The pivot inverts both the architecture (local) and the customer (single individual).
- Competitive landscape: SwingVision (\$179.99/yr, cloud, generic model, 200k+ users), BadPro+ (pay-per-use), Dartfish/Hudl (coach/team, cloud, manual tagging). None offer a **per-user self-optimizing local model**. Edge-AI/privacy is a real and growing trend but mostly in surveillance, not consumer sports apps. Subscription fatigue is pushing toward one-time + paid-upgrade models.
- The pivot's mechanism (LOVO harness + correction loop as per-user labels + per-user cue selection) is a genuine engineering asset, but it creates product tensions: cold-start (needs 10-15 videos), the "batteries upgrade" reduces personalization's differentiator, and per-user overfitting weakens the network-effect/data moat.

16. user (2026-06-14T08:26:40.809Z)

Structured output provided successfully

17. assistant (2026-06-14T08:26:55.049Z)

I've completed the product/PMF/market analysis of the personalization-first pivot and returned it via StructuredOutput.

Summary of what I grounded the analysis in:

Internal docs read: ``docs/PMF_MARKET_RESEARCH.md``, ``docs/COMMERCIALIZATION.md``, and ``docs/EXPERIMENT_HARNESS.md`` ? which together establish that the project's **existing** market-of-record assumes the **opposite** thesis (cloud-hosted SaaS, generalization-first, India badminton **academies** as the #1 B2B beachhead, chosen precisely because individual amateurs are low-WTP/high-churn). The harness doc confirms the per-user mechanism the pivot relies on (``compare()``/``compare_unlabeled()``, LOVO, per-cue enable flags, correction-loop labels) genuinely exists ? the owner's claim that it was designed to enable this is correct at the engineering layer.

Web research (4 searches): competitive landscape is uniformly **cloud + generic-model** (SwingVision \$179.99/yr cloud, 200k+ users; BadPro+ pay-per-use; Dartfish/Hudl coach-facing cloud) ? no incumbent ships a per-user, self-optimizing, on-device model, so the pivot targets real whitespace. Edge/on-device AI is a fast-growing, privacy-driven trend (~\$25B-\$119B) but proven in surveillance, not consumer sports. Subscription fatigue is reviving one-time/perpetual licenses, supporting the proposed "one-time install + batteries-upgrade sub" hybrid.

Core tensions surfaced: the cold-start chasm (value needs 10-15 videos vs incumbents' instant first-clip output), the "batteries upgrade" partially cannibalizing the personalization differentiator, and the pivot trading the cross-user **data flywheel** the strategy docs call THE moat for a per-user **switching-cost** moat ? which reframes this as a premium-tool business rather than a platform business, with the corpus-rich 100+hr professional (not the casual hobbyist) as the right beachhead.

The full pros/cons/findings/shapes-future/open-questions are in the structured output.

Sources:

- [SwingVision pricing/features](https://swing.vision/subscribe)
- [Dartfish vs Hudl comparison](https://toolsinfo.com/compare/dartfish-vs-hudl)
- [On-Device AI and Edge 2025](https://blog.madrigan.com/en/blog/202605131630/)
- [MatchMotion / badminton AI analysis](https://www.researchgate.net/publication/404597971_MATCHMOTION_AI-POWERED_MOBILE_APPLICATION_FOR_AUTOMATED_BADMINTON_VIDEO_ANALYSIS_USING_YOLOV8_OBJECT_DETECTION)
- [Software licensing: subscription vs one-time 2025](https://arturo.winandoffice.com/the-future-of-software-licensing-subscription-vs-one-time-purchase-in-2025/)